MILITARY SPECIFICATION

RECORDING GROUP, VIDEO SIGNAL
OA-8962/ASH

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall provide a 16 mm film recording of the display data emanating from the infrared Detecting System AN/AAS-36 or a Scan Converted Radar.

1.2 Classification. The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Recorder, Video Signal RO-515/ASH	3.5.1
Control, Recorder C-10522/ASH	3.5.2

1.3 Associated Equipment. This equipment shall operate with the associated equipment listed in 6.6.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-5088	- Wiring, Aerospace Vehicle
MIL-E-5400	- Electronic Equipment, Airborne, General Specification for
MIL-T-5422	- Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	- Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	- Panels, Information, Integrally Illuminated
MIL-M-7793	- Meter, Time Totalizing
MIL-E-17555	- Electronic And Electrical Equipment Accessories, And Repair Parts, Packaging and Packing Of
MIL-T-18303	- Test Procedures, Preproduction, Acceptance And Life for Aircraft Electronic Equipment, Format For
MIL-N-18307	- Nomenclature And Identification For Electronic, Aeronautical, And Aeronnautical Support Equipment Including Ground Support Equipment
MIL-C-81511	- Connector Electrical, Circular, High Density, Quick Disconnect, Environment Resisting, And Accessories, General Specifications For

STANDARDS

FEDERAL

FED-STD-3	- Color, Aeronautical Lighting
FED-STD-595	- Colors

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MILITARY

MIL-STD-461 - Electromagnetic, Emission And Susceptibility, Requirements For The Control of Electromagnetic Interference

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of

MIL-STD-704 - Aircraft Electric Power, Characteristics

MIL-STD-781 - Reliability Test Exponential Disbribution

MIL-STD-785 - Reliability Program For Systems And Equipment Development And Production

MIL-STD-794 - Parts and Equipment, Procedures For Packaging And Packing Of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Electronic Industries Association (EIA)

Standard RS-343-A - Electrical Performance Standards for High Resolution Closed Circuit Television Camera

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, D.C. 20006.)

American National Standards Institute (ANSI)

PH22.7-1964 - 16 mm Motion Picture Aperture Image

PH22.110-1965 - 16 mm Motion Picture Film, 2R-2994

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

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3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.1.1 First article. When specified (See 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (See 4.1.1 and 6.3).

3.2 Parts and material. All parts and material shall be in accordance with MIL-E-5400. Microelectronic technology shall be considered and microelectronic devices shall be given first consideration in design.

3.2.1 Non-repairable assemblies. Use of non-repairable assemblies shall be in accordance with MIL-E-5400.

3.2.2 Non-standard parts and materials approval. Approval for use of non-standard parts and materials shall be in accordance with MIL-E-5400.

3.3 Design and construction. The equipment shall be in accordance with MIL-E-5400, Class I Equipment, for design, construction and workmanship.

3.3.1 Total weight. The total weight of the equipment, excluding film, shall be not greater than 50 pounds.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued.

3.3.2.1 Operational stability. The equipment shall operate continuously or intermittently for a period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have a total operating life of 13,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in mean time between failure (MTBF). The equipment, including Built-in-Test provisions, shall have 1000 hours of a mean (operating) time between failures when tested and accepted in accordance with 4.4.2.

3.3.2.4 Time totalizing meter. The following unit shall contain a time totalizing meter in accordance with MIL-M-7793.

Unit	Type of Meter
Recorder, Video Signal	MS 17322

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3.3.3 Cabling and connections. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.1 External connectors. The external power and control signal connectors shall be in accordance with MIL-C-81511. One connector shall be located on the rear of the control unit and one on the front panel of the recorder unit. Two separate triax connectors, which are not interchangeable, shall be provided on the front panel of the recorder unit for input video connections.

3.3.3.2 Interconnecting cabling. The equipment shall be capable of operation when interconnected by aircraft wiring in accordance with MIL-W-5088. External cables and that portion of the connectors attached to the external cables shall not be supplied as part of the equipment.

3.3.4 Control panels. The equipment's control panels shall be in accordance with MIL-C-6781.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Interference control. The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to electromagnetic interference shall be in accordance with MIL-STD-461 when measured in accordance with MIL-STD-462, except as modified herein:

a. 400 Hz power lines.

CE-01: MIL-STD-461, Figure 13. 30 dB relaxation for 400 Hz and odd multiples to and including the 7th harmonic for three-phase Y or delta. 20 dB relaxation (Figure 13) for all other multiples of 400 Hz (narrowband and broadband measurements).

b. 28 Volt power lines.

CE-01: MIL-STD-461, Figure 13. 20 dB relaxation for all short-duration pulses at low prf (less than 100 Hz) and step functions provided "RE" requirements are met by proper cabling.

CE-03: MIL-STD-461, Figure 14. 20 dB relaxation at frequencies up to and including 2 MHz where the primary power smoothing filter is a low-impedance shunt (less than one ohm).

c. Control and signal lines.

CE-02; 04: MIL-STD-462, paragraph 5, note C2. Change From: "in no case shall a group of leads be probed with both the high and return lead in the test group, except for twisted pairs." To: "Wires may be tested in a bundle or group (pairs, triplets, quads, etc.) as they will be installed."

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d. System.

RE-01: MIL-STD-461, Figure 20. Slope limit. Change From: "-40 dB per decade." To: "-20 dB per decade starting from present origin at 30 Hz." (60 dB relaxation at 30 KHz).

e. System.

RS-02: MIL-STD-461 paragraph 6.18(a) (case test only) Change From: "20 amperes applied to test wire at the power frequency(ies)." To: "5 amperes applied to test wire at the power frequency(ies)." (12 dB relaxation at 400 Hz).

3.3.7 Provisions for maintainability. Built-In Test features, construction, provisions for test points, and other maintainability parameters shall be as specified herein.

3.3.7.1 Compatibility with VAST. Equipment compatibility with the Versatile Avionic Shop Test system (VAST) shall not be required.

3.3.7.2 Maintenance panel. A separate maintenance panel shall not be required.

3.3.7.3 Fault indicators. In-flight performance monitoring and fault indication shall be provided by a fault indicator located on the front panel of the recorder unit and by the recording indicator light located on the control unit.

3.3.8 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The conditions shown in Table I shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

TABLE I. Standard conditions.

Environment	Condition
Temperature	Room ambinet (25°C ± 10°C)
Alitutde	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 + 1.0 VAC 400 Hzs and 26.5 ± 0.5 VDC

3.3.10 Service conditions. The equipment shall operate under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 1 equipment and outlined in Table II.

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TABLE II. Service conditions.

Environment	Condition
Altitude	Atmosphere to 50,000 ft (29.9 to 5.56 in Hg)
High Temperature	+71°C (160°F)
Low Temperature	-54°C (-65°F)
Temperature Shock	+71°C to -54°C within 5 min
Temperature Altitude	+71°C to -54°C Atmosphere to 50,000 ft
Humidity	Up to 100% relative humidity, including condensation and frost
Salt Fog	Exposure to salt-saturated air 0.2 grams /m^2 of liquid water
Explosive Atmosphere	Shall not cause an explosion when operated in explosive atmosphere
Acceleration (Linear)	Fore and Aft 5 g's peak Up 8.5 g's peak Down 12 g's peak Lateral ±3 g's peak
Vibration	In accordance with MIL-E-5400 Figure 3, Curve 1
Shock (design objective)	15 g's peak, 11 ± millisecond in each of three planes

3.3.11 Warm-up time. The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall be not greater than 3 minutes under standard conditions and 10 minutes after storage at -20°C. Warm-up time for the equipment between -20°C and -54°C shall be not greater than 20 minutes.

3.3.12 Input electrical power. The electrical power inputs for the equipment shall be as follows:

3.3.12.1 Operating power. The equipment shall meet all applicable requirements and shall give specified performance when energized from the following power sources having characteristics and limits in accordance with MIL-STD-704. The power required shall be not greater than the specified amounts.

a. AC power (three phase) 115V line to neutral, 400 Hz, balanced "Y" Category "B", 350 VA maximum while heating; 250 VA in Record Mode.

b. DC Power: +28 VDC, 0.2 Amp, isolated from system ground.

3.3.12.3 Lighting power. Input power for lighting shall require not more than 0.5 amps at 28 volts.

3.3.12.3 Degraded performance. Degraded performance shall be permitted during no-power periods per Curve 3, Figure 3 of MIL-STD-704. Upon restoration of normal power the equipment shall resume normal operation.

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3.3.12.4 Standby power. The equipment shall provide a standby mode whereby rated voltage is applied to the CRT filament, and at temperatures below 50°F, to the magazine heater. The standby mode shall be activated with the application of aircraft power and the control unit's record select switch set to either the 1 FPS or 10 FPS positions.

3.3.13 Cooling. The equipment shall not require external cooling air for operation under standard or service conditions.

3.3.14 Color. The color of the external surfaces of the recorder unit shall be Beige as specified in FED-STD-595, Color No. 26555.

3.3.15 Finish. The finish on the equipment shall be resistant to corrosion. All burrs, sharp edges, and sharp corners shall be removed prior to final finish.

3.4 Performance. Unless otherwise specified herein, the values set forth in this specification shall establish the requirements for performance and shall apply to performance under the standard conditions specified in 3.3.9 and under the service conditions specified in 3.3.10.

3.4.1 Operation. The equipment shall record output video display signals from either the AN/AAS-36 Infrared Detecting System or a Scan Converted Radar, but not simultaneously, on standard 16 mm film. The control unit provides remote operation of the recorder unit. Photographic recordings are made at film frame rates of either ten frames per second or one frame per second, selectable from the control unit by a record select switch.

3.4.2 Interface characteristics. The received signal shall be a composite video output from the AN/AAS-36 Infrared Detecting System, or a Scan Converted Radar. The composite video signal format shall be a standard 875/60 television line signal in accordance with RS-343. The TV frame rate and scan timing shall be established and controlled by the internal timing circuits in the AN/AAS-36 System or Scan Converted Radar. The video bandwidth during active line time shall be 10 MHz.

3.4.2.1 Composite video. The received composite video signal shall have the characteristics shown in Table III in accordance with RS-343.

TABLE III. Composite video characteristics.

Video signal	Characteristics
Polarity	White level positive
Amplitude	0.7 volts peak to peak when terminated into 75 ohms
Bandwidth	DC to 10 MHz
Roster (Television)	875 lines interlaced 2:1
Field Rate	60 Hz
Frame Rate	30 Hz

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3.4.2.2 Aspect ratio. The aspect ratio of the displayed Infrared Detecting System scene shall be 4 units azimuth by 3 units elevation (4:3). The aspect ratio of the Scan Converted Radar display scene shall be 1 unit azimuth by 1 unit elevation (1:1).

3.4.2.3 Impedance. Recorder input impedance shall be 75 ohms.

3.4.2.4 Interface cabling. The video interface cable shall be 75 ohm triax. The cable outer shield termination shall be grounded to the recorder chassis.

3.4.2.5 AN/AAS-36 generated gray scale. A reference gray scale pattern is selectable on the AN/AAS-36 System. This reference pattern consists of a minimum of eight shades of gray presented at the bottom of the display roster. When selected, this gray scale signal will be recorded and will override the automatic contrast control of the recorder.

3.4.3 Performance characteristics. The recorder shall meet the following performance characteristics:

3.4.3.1 Resolution. For an input video signal which generates three bar test patterns both vertical and horizontal on the CRT face, the recorder shall provide not less than the following per cent modulation:

a. 70% modulation at 20 line pairs/mm

b. 80% modulation at 10 line pairs/mm

This modulation shall be measured off the developed film. Development of the film shall be under standard processing conditions. (Negative processing.)

3.4.3.2 Geometrical distortion. The recorder shall not add more than 2 percent geometrical distortion in linearity to the developed picture.

3.4.3.3 Automatic Gain Control (AGC). The recorder shall provide automatic gain control to maintain a constant video level independent of variation of incoming video level, within standard specified limit, in accordance with RS-343.

3.4.3.4 Automatic Contrast Control (ACC). The recorder shall provide an automatic contrast control, consisting of a peak/average contrast sensor, in which the display output is sensed and negative feedback is applied. The automatic contrast control shall provide full output contrast (eight gray shades) for an input video contrast of two gray shades.

3.4.3.5 Gamma. The transfer characteristics of the film, relating log brightness vs film density, shall not deviate from a least squares line by more than 0.08 in density. The slope of the least squares straight line shall lie between 1 and 1.3 for an input gamma of 0.55.

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3.4.3.6 Bandwidth. Bandwidth of the television system, measured at the cathode of the display tube, shall be sufficient to meet the resolution requirement, but shall not be less than 10 MHz. Bandwidth is defined as the frequency at which the signal is 3 dB lower than a reference signal at 1 MHz.

3.4.3.7 Signal-to-noise ratio. The input video may vary from a signal-to-noise ratio of 3 to 1 up to 90 to 1. In the presence of pulse signals, occupying 3 percent of the picture area and having an amplitude up to four times the normal input signals, normal signal shall be recorded with flare not greater than 6 percent of the picture area.

3.4.3.8 Gray scale. The recorder shall reproduce a minimum of eight standard gray shades (contrast ratio of 11.3 to 1) for a video input signal. The recorder shall also contain an internal gray scale generator which provides a gray scale pattern on the CRT display when the gray scale switch on the recorder unit, or the test pattern switch on the control unit, is activated.

3.4.4 Camera characteristics. The camera shall be a pulse operated type, synchronized with the video display, with a fixed lens system focused on the CRT display.

3.4.4.1 Film size. The camera shall use 16 mm motion picture film, perforated along both edges with a pitch of 0.2994 inch, in accordance with PH22.110.

3.4.4.2 Image format. The camera aperture image area on the film shall be nominally 0.404 by 0.295 inches in accordance with PH22.7.

3.4.4.3 Film rate. The camera shall provide two selectable film frame rates, one at ten frames per second and the other at one frame per second. Frame exposure shall be synchronized with the display.

3.4.4.4 Film magazine. The camera shall include an easily replaceable film magazine. The magazine shall have a coaxial configuration of the supply and takeup spools.

3.4.4.5 Film capacity. The magazine shall have maximum film capacity of not less than 450 feet of thin base, 0.004 inch, film (i.e. Kodak RAR 2498 or equal).

3.4.4.6 Film remaining indicator. A film remaining indicator located on the control unit shall numerically display in feet (± 1 foot) the length of the remaining unexposed film in the magazine.

3.4.5 Recorder control logic. The recorder shall include the required control logic for controlling the camera motor, film advance mechanism, CRT circuitry and power supplies in accordance with the selected functional modes as follows:

a. Operational. The recorder control logic shall interface externally with the remote control unit.

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b. Testing. The recorder control logic shall internally interface with its front panel controls, enabling these control functions only when the remote control unit is disconnected from the recorder.

3.4.6 Protective features. The following protective features shall be incorporated in the recorder. These protective features shall be interfaced with the BITE system as described herein:

a. Low input line voltage. Whenever the 3 phase 115 VAC, 400 Hz input voltage drops below 108 VAC, the recorder shall be inhibited from recording. The RECORD indicator lamp shall go out, but the BITE indicator shall not set a fault.

b. No video input. If the recorder should be selected to record, and no composite video signal is present at the selected input, the RECORD indicator shall not light, and the BITE indicator shall not set a fault.

c. Sweep failure. Failure of either the horizontal or vertical sweep, or both, shall remove the high voltage from the CRT. The RECORD indicator shall not light and a BITE fault indication shall be set.

3.4.7 Built-In-Test Equipment (BITE). The recorder shall include BITE circuitry to provide an indication of the operational status of the equipment. A BITE fault indicator shall be located on the front panel of the recorder unit.

3.4.7.1 Fault conditions. The following conditions shall be considered an equipment failure which individually or collectively shall set a BITE fault indication.

a. No high voltage

b. Loss of a low voltage power supply

c. No film motion in record mode

d. Unblanking of CRT defective

e. No CRT filament

f. Nonsynchronous sweep

g. Loss of scans

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3.4.7.2 Implementation. Upon detection of a failure, the BITE indicator shall be set and latched. The indicator shall remain latched in the fault position until deliberately reset. The BITE indicator shall be resettable by use of a small magnet or the electrical reset coils. The BITE indicator shall only be actuated by deliberate man-initiated action (e.g. after bench test station repairs).

3.5 Detail requirements. The video signal recording group shall consist of two units; a video signal recorder unit and recorder control unit. When installed in the aircraft, the two units shall be interconnected by aircraft wiring. Interconnecting cabling shall not be part of the equipment supplied.

3.5.1 Recorder, video signal RO 515/ASH

3.5.1.1 Function. The video recorder unit shall receive input power and a composite video signal from the aircraft. The recorder shall convert the received video signal into a CRT display and photographically record the displayed imagery on 16 mm motion-picture films. The recorder shall include: the necessary signal processing, CRT display and control circuitry with associated power supplies, the required optics and camera mechanism, a removable film magazine, and include built-in test features. The recorder shall be remotely operated by the control unit.

3.5.1.2 Form factor. The recorder unit shall be enclosed in a case 17.00 inches in length, 13.00 inches in width, and 8.72 inches in height, and configured as shown in Figures 1 and 2.

3.5.1.3 Weight. The weight of the recorder unit shall be not greater than 48 pounds, excluding film.

3.5.1.4 Controls. The recorder unit shall contain the following maintenance controls located on the front panel as shown in Figure 1. These controls shall be interfaced with the remote controls, such that when connection is made to the control unit, these controls shall be inoperable except for the CAMERA TEST switch.

a. 1 FPS/PWR OFF/10 FPS. A three position toggle switch, center off, two latch positions. The switch shall permit the selection of either 1 or 10 frames per second (FPS) recording rates, and shall apply input power to the recorder in either position.

b. RECORD START/STOP. A two position toggle switch. The switch shall start or stop the normal operation of the recorder.

c. FLIR/RADAR. A two position toggle switch. The switch shall provide for the selection of either the FLIR or RADAR composite video inputs to the recorder unit.

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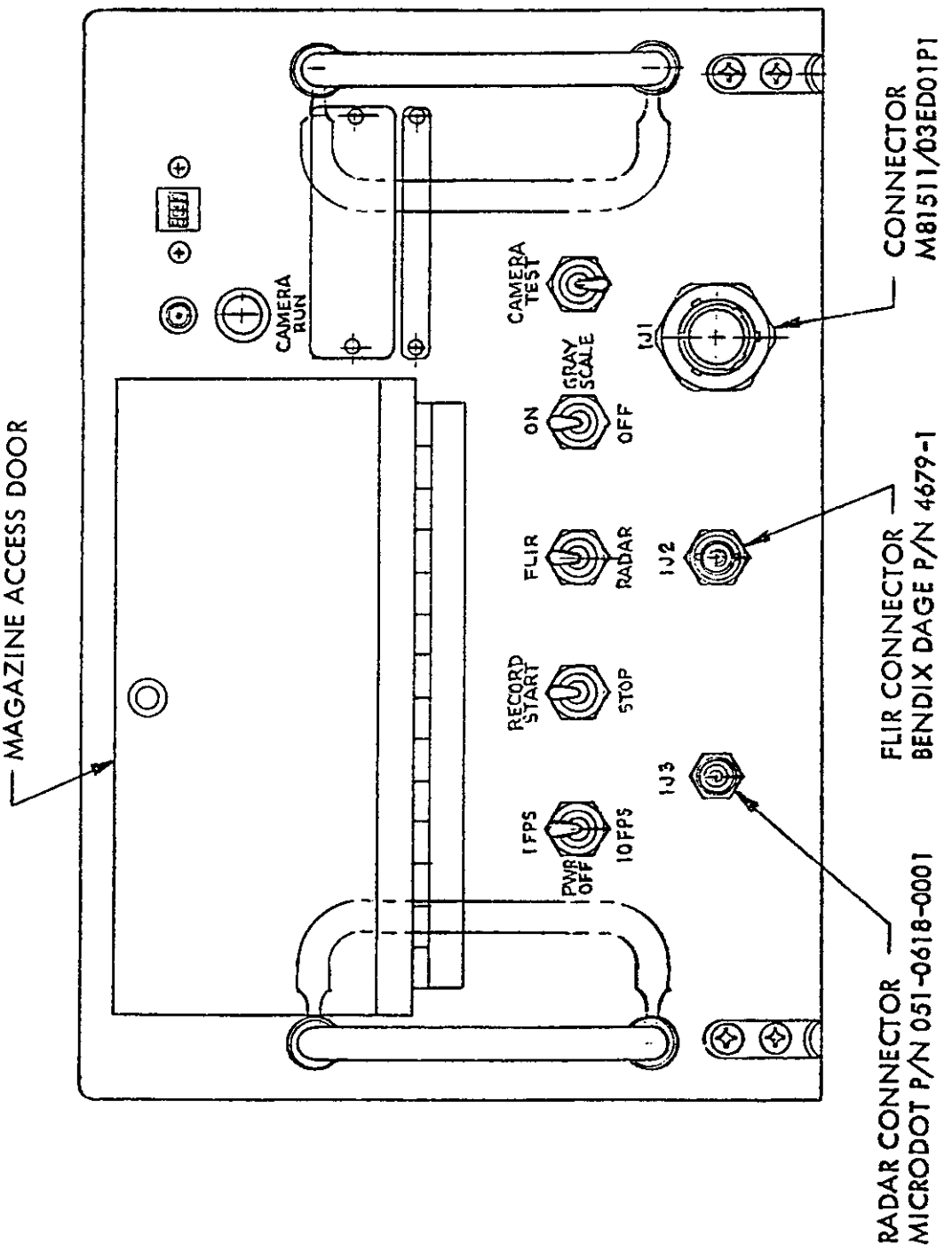


FIGURE 1. Recorder unit front panel and controls.

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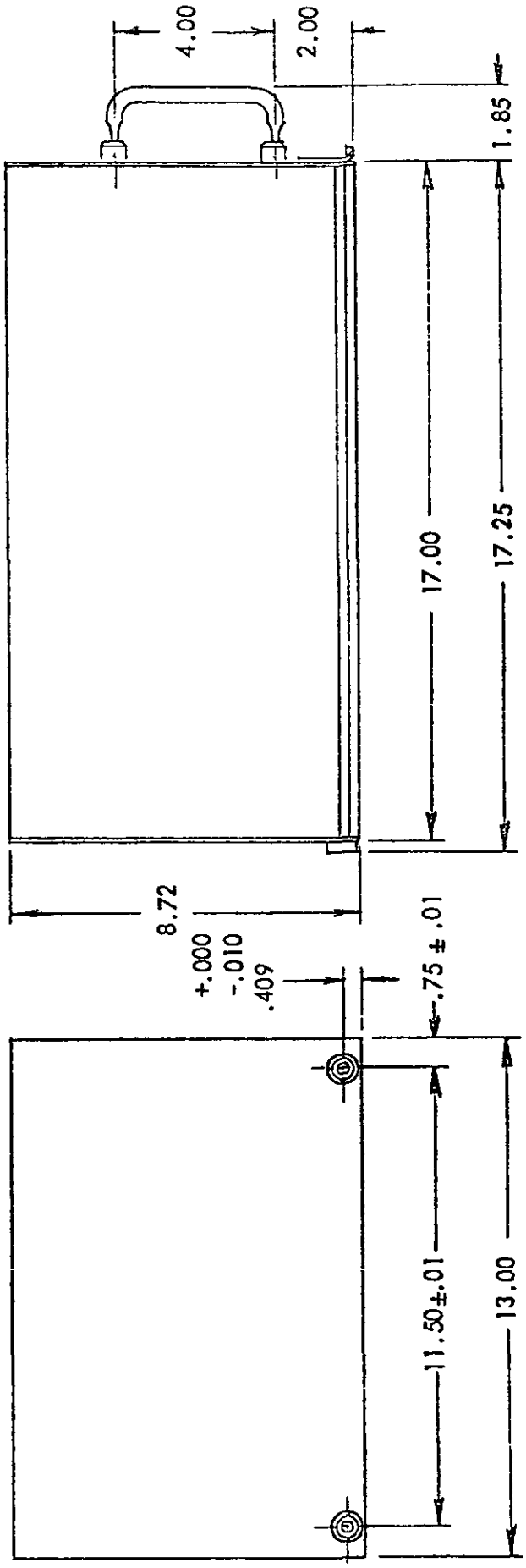


FIGURE 2. Recorder unit outline.

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d. GRAY SCALE ON/OFF. A two position toggle switch. The switch shall control the displaying of an internally generated gray scale pattern on CRT.

e. CAMERA TEST. A two position, spring loaded, toggle switch. The switch shall provide momentary contact control to test the recorder's film transport mechanism, to take up film after magazine replacement, and verify proper operation prior to flight. The switch shall only be used to operate the recorder when no video or sync signals are applied to the recorder.

f. CAMERA RUN. The camera run indicator shall light whenever the recorder unit is recording on film and operating normally. The indicator shall be interfaced with the BITE circuitry and shall not light if a BITE fault is indicated. The CAMERA RUN light shall be green in color as specified in FED-STD-3.

g. BITE Indicator. The BITE indicator shall be a magnetic latch type. The indicator shall remain latched in the fault position (white face) until deliberately reset.

h. Elapsed time indicator. The elapsed time indicator shall be a digital readout electromechanical clock timer indicating, in hours, the cumulative operating time of the recorder.

3.5.1.5 Electrical Connections. Connections to the aircraft interconnecting wiring shall be as shown in Table IV.

TABLE IV. Recorder unit electrical connectors.

Reference Designation	Receptacle Type	Function
1J1	M81511/03ED01P1	Input power and interface to the control unit
1J2	BENDIX DAGE PN 4679-1	Input video signal from Infrared Detecting System
1J3	MICRODOT PN 051-0618-0001	Input video signal from Scan Converted Radar

3.5.1.6 Contents. The recorder unit shall be classified a Weapons Replaceable Assembly (WRA), and consist of fifteen major subassemblies classified as Shop Replaceable Assemblies (SRA). The subassemblies shall include: ten standard sized plug-in circuit cards, two power units, an electro-optical assembly, the film magazine and the basic housing unit, as shown in Table V.

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TABLE V. Shop replaceable assemblies.

Assembly Ref. Design.	Description
1A1	Printed circuit card assembly, consisting of the gamma control functional circuitry
1A2	Printed circuit card assembly, consisting of the auto contrast functional circuitry
1A3	Printed circuit card assembly, consisting of the video switch functional circuitry
1A4	Printed circuit card assembly, consisting of the horizontal blanking and gray scale generator functional circuitry
1A5	Printed circuit card assembly, consisting of the vertical blanking functional circuitry
1A6	Printed circuit card assembly, consisting of the unblank and sync regenerator functional circuitry
1A7	Printed circuit card assembly, consisting of the vertical scan functional circuitry
1A8	Printed circuit card assembly, consisting of the motor control functional circuitry
1A9	Printed circuit card assembly, consisting of the video built-in test functional circuitry
1A10	Printed circuit card assembly, consisting of the general built-in test functional circuitry
1A11	High voltage power supply unit, which shall provide the following voltages: +15 KVdc ±5%, 30 microamp +3 to 5 KVdc adjustable, 300 microamp +500 Vdc ±5%, 50 microamp
1A12	Low voltage power supply unit, which shall provide the following voltages: +100 Vdc ±1%, 0.1 amp +20 Vdc ±0.6 V, 1.5 amp +28 Vdc ±1.0 V, 0.3 amp +15 Vdc ±1%, 2.0 amp -15 Vdc ±1%, 1.0 amp + 5 Vdc ±5%, 2.0 amp - 5 Vdc ±5%, 0.2 amp +28 Vdc ±10%, 3.0 amp (Unregulated) + 6.3 Vdc ±5%, 0.3 amp
1A13	Electro-optical assembly, which shall include the CRT display, the camera and the required optics. The assembly shall consist of an optical aligning base plate and contain the following replaceable items: a camera sub-assembly, the CRT and yoke subassemblies,

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TABLE V. Shop replaceable assemblies. (Continued)

Assembly Ref. Design.	Description
1A13 (Contd)	two mirrors and a sensor subassembly, and the video/deflection drive circuitry contained on three replaceable printed circuit cards.
A15	Film magazine assembly, which shall contain the required film transport mechanism, film storage, film remain indications and a heater. The magazine shall be a self-contained, quick replaceable unit, which automatically aligns and interfaces with the camera body when installed. The film magazine shall be classified as an SRA for intermediate maintenance repairs and as a WRA for operational maintenance (e.g., replacement of exposed film).

3.5.2 Control recorder C-10522/ASH. The control unit shall be in accordance with MIL-C-6781 for Type II control panel. The lighting of the plastic panel shall be in accordance with MIL-P-7788, Class 2W, Type V, with a lamp voltage of 28 volts. Panel markings shall be as follows, and as shown in Figure 3.

a. Letters and numbers shall be Gorton Normal.

b. Letter height shall be interpreted as the overall height of letter face.

c. Panel title shall be 18-point lettering (0.18 ± 0.005 inch high).

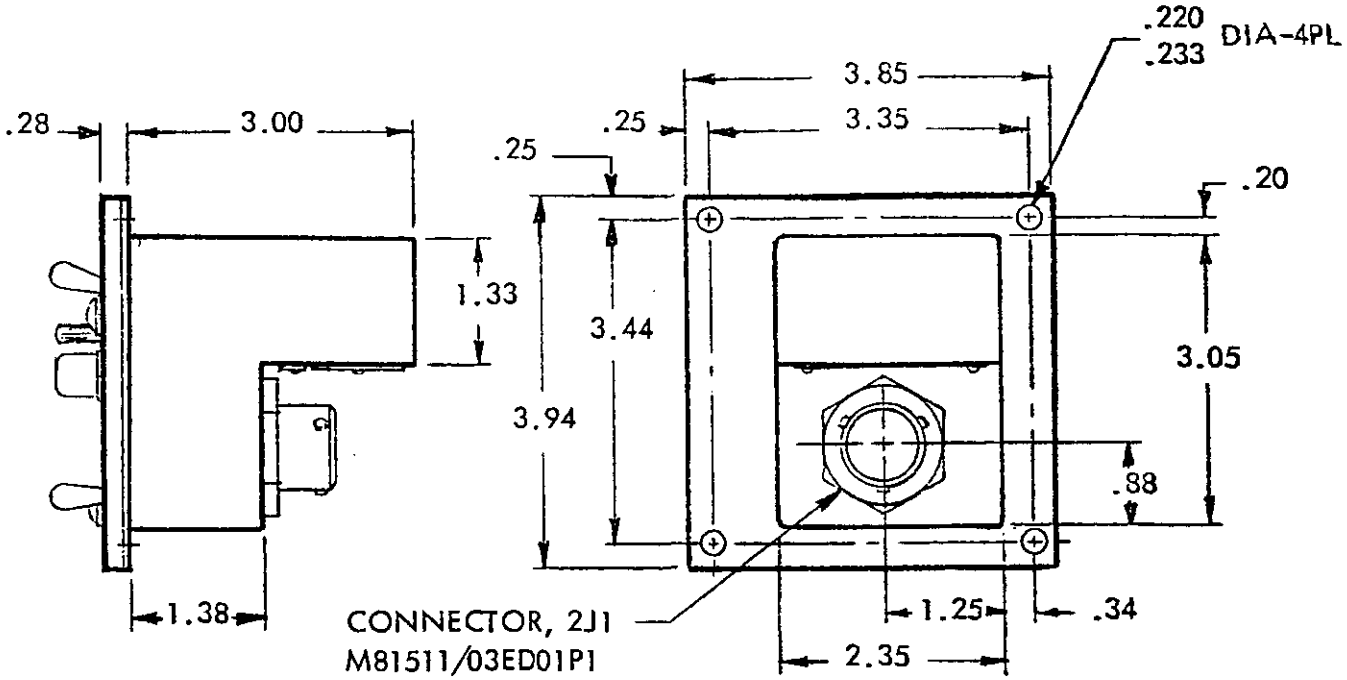
d. Control titles shall be 16-point lettering (0.16 ± 0.005 inch high).

e. Control positions shall be 12-point lettering (0.12 ± 0.005 inch high).

3.5.2.1 Form factor. The control unit shall be enclosed in a case with the front panel 3.94 inches in height, 3.85 inches in width, and a case depth of 3.00 inches and configured as shown in Figure 3.

3.5.2.2 Controls. The control panel shall have the following controls located as shown in Figure 3.

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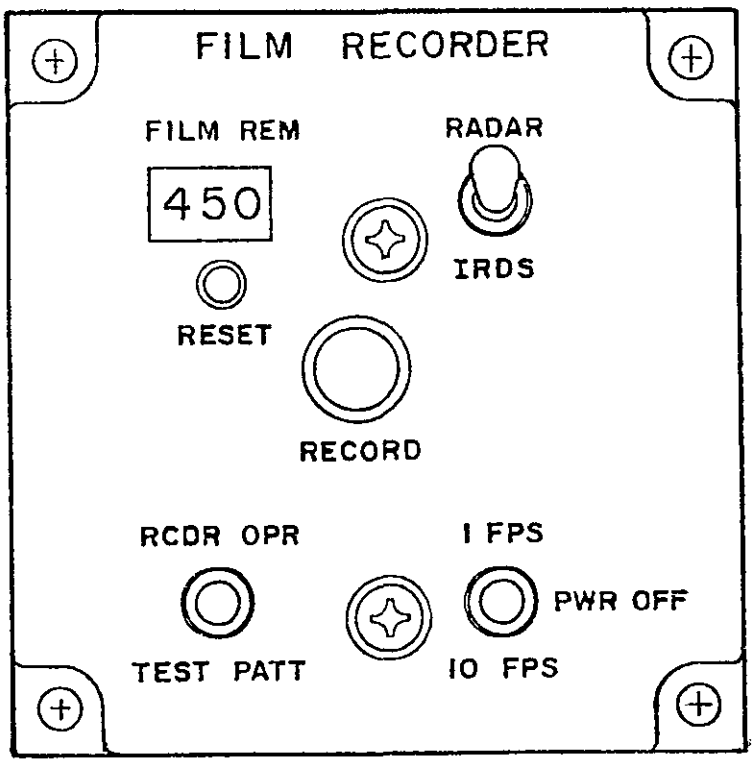


FIGURE 3. Control unit outline and controls.

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a. IRDS/RADAR. A two position toggle switch. The switch shall permit selection of either the Infrared Detecting System (IRDS) or Scan Converted Radar (RADAR) composite video signal to be recorded.

b. 1 FPS/PWR OFF/10 FPS. - A three position toggle switch, center off, two latch position. The switch shall permit the selection of either the 1 or 10 frame per second (FPS) recording rates, and shall apply input power to the recorder in either position.

c. RCDR OPR/TEST PATT. A three position toggle switch center off, one momentary and one latch position. The RCDR OPER (latch) position shall cause the recorder to operate, recording either the IRDS or RADAR imagery, as selected by the IRDS/RADAR switch. The TEST PATT. (momentary) position shall cause the recorder to operate, recording the internally generated gray scale, when the mixed sync signal is received from the system selected by the IRDS/RADAR switch.

c. FILM REM. An electromechanical countdown indicator, with a digital readout in feet, which indicates the amount of unexposed film remaining in the magazine. The indicator shall have a manual RESET for resetting the indicator whenever a magazine is replaced.

d. RECORD. The recording indicator shall light whenever the recorder unit is recording and operating normally. The recording indicator shall be interfaced with the BITE circuitry and shall not light if a BITE fault is indicated on the recorder. The RECORD light shall be green in color, as specified in FED-STD-3.

3.5.2.3 Electrical connection. Connection to the aircraft interconnecting wiring shall be as shown in Table VI

TABLE VI. Control unit electrical connector.

Reference Designation	Receptacle Type	Function
2J1	M81511/03ED01P1	Interface connection to recorder unit and aircraft panel lighting power

3.5.2.4 Parallel aircraft function. The control unit shall provide two contacts in its interface connector (2J1) which are in parallel with the RCDR OPR switch. These contacts may be used for an external switch employed in the aircraft to cause the recorder to operate, recording imagery selected by the IRDS/RADAR switch.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. Items covered by this specification shall be subjected to the following tests as required by the contract. (See 6.2)

a. First article (See 4.2)

b. Initial production tests (See 4.3)

c. Acceptance tests (See 4.4)

d. Life tests (See 4.5)

4.1.2 Inspection conditions. Unless otherwise specified in the contract, all inspection and tests shall be performed in accordance with conditions specified herein. When approved test procedures are available from previous contracts, such procedures may be used. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.2 First article tests. First article tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. (See 6.3) First article tests shall be in accordance with the government approved test procedure. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness the tests when so desired.

4.2.1 Scope of tests. First article tests shall demonstrate that the equipment supplied shall meet the requirements of this specification, applicable subsidiary specifications, and the contract. The first article testing shall include performance tests under standard conditions and limited environmental testing. The environmental testing shall be conducted under the simulated environmental conditions shown in Table VII in accordance with the procedures set forth in MIL-T-5422.

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TABLE VII. Simulated environmental conditions.

Environment	Condition
Temperature	-20°C to +85°C ±5°C
Altitude	Atmospheric to 50,000 ft ±5%
Vibration	0.01 in DA to 60 Hz ±10% ±2G 60 Hz to 500 Hz ±10%

4.2.2 First article test data. The contractor shall maintain all data collected in conducting these tests for Government inspector review.

4.2.3 First article approval. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all tests (See 6.3). No production equipments shall be delivered prior to the approval of the first article sample. The approved first article sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance.

4.2.4 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection of the equipment and for equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the Government inspector.

4.3 Initial production tests. When required by the contract (See 6.2) one of the first ten production equipment shall be selected and sent at the procuring activity's expense to a designated Government laboratory for tests. This equipment shall be selected by the Government inspector after the equipment has successfully passed all individual tests. The first article sample shall not be selected for this test.

4.4 Acceptance tests. The contractor shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. The contractor shall maintain test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor. Acceptance tests shall consist of the following:

a. Individual tests (See 4.4.1)

b. Reliability assurance tests (See 4.2.2)

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4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of product (See 4.4.1.1)

b. Operational test (See 4.4.1.2)

c. Photographic test (See 4.4.1.3)

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure equipment operation.

4.4.1.3 Photographic test. Each equipment shall be operated exposing a test film and the film processed and evaluated to assure photographic recording.

4.4.1.4 Test conditions. The individual tests shall be conducted under standard laboratory conditions.

4.4.1.5 Test detail. The test details such as performance characteristics to be measured, normal maintenance action (i.e., film reloading) failure criteria, preventive maintenance allowed, etc., shall be in accordance with the approved test procedure. (See 6.2.2)

4.4.2 Reliability assurance test. A production reliability assurance, or burn-in, test shall be conducted on each equipment submitted for acceptance. The tests shall consist of fifteen temperature cycles in accordance with MIL-STD-781, test level E, except as modified herein.

4.4.2.1 Scope of test. Each equipment shall be operated under the conditions specified herein for a period of 50 hours. The equipment shall be operated during the heating cycle and switched off during the cooling cycle. Vibration shall be applied to the equipment for a period of 10 minutes during each hour of operation. A simulated film load may be used, in lieu of photographic film, for camera operating during the testing.

4.4.2.2 Test conditions. The tests shall be conducted under the simulated service condition shown in Table VIII.

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TABLE VIII. Simulated service condition.

Test	Condition
Input Power	115 VAC ±5 VAC, 380 to 420 Hz 26.5 Vdc nominal, 22 to 28 Vdc
Temperature/ Cycle Time	-54°C ±2°C/2 hours +55°C ±2°C/3 hours
Vibration	2.2G ±2G peak 10 to 60 Hertzs

4.4.2.3 Test details. The test details such as performance characteristics to be measured, failure criteria, preventive maintenance allowed, etc. shall be in accordance with the approved test procedure. (See 6.2.2)

4.5 Life test. When required by the contract (See 6.2), the contractor shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test sample shall be selected by the Government inspector in accordance with Table IX. Equipments which have successfully passed the Initial Production Test may be selected for life tests. Test time accumulated during the Reliability Assurance Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment.

TABLE IX. Life test sampling.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1 for each additional 500 or fraction thereof

4.6.1 Test conditions. The life test shall be conducted under the conditions shown in Table X.

TABLE X. Life test conditions.

Condition	Limits
Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ±5 volts (at lowest applicable frequency)
D.C. Voltage	27.5 ±2.0 volts

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4.6.2 Test periods. The test may be run continously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.6.3 Performance check. Performance checks shall be made at eight hour intervals throughout the test period.

4.6.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.7 Equipment failure. Should a failure occur during either the individual or reliability assurance tests, the following action shall be taken. Classification of a failure shall be in accordance with MIL-STD-781.

a. Determine the cause of failure

b. Determine if the failure is a random or pattern failure

c. If the failure is determined to be a relevant pattern failure, the proposed corrective action intended to reduce the possibility of the same failure occurance shall be submitted to the Government inspector for approval.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.7.1 Failure report. In the event of a failure, the Government inspector shall be notified immediately and a failure report made upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which

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occurred. The procuring activity will review such proposals and determine whether they are acceptable. (See 6.2.2)

4.8 Reconditioning of tested equipment. Equipment which has been subjected to first article, initial production of life tests may be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.9 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

4.10 Rejection and retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order (See 6.2) in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. This equipment is intended for use with the P-3C aircraft and shall record on 16 mm photographic film the CRT display of composite video signal provided to its inputs, either from the AN/AAS-36 Infrared Detecting System, or a Scan Converted Radar.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification

b. Testing requirements (See 4.1.1)

c. Selection of applicable levels of packaging and packing (See 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7.104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved

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Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirement List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.1.5 and 4.4.2.3	Procedures, Test	UDI-T-21347
4.7.1	Report, Failure	UDI-R-21141

(Copies of data item descriptions required by the contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulations. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any specification.

b. This specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.5 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and in improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result, it is desired they be investigated. When investigation shows advantages can be realized, an Engineering Change Proposal (ECP) shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 Associated equipment.

a. Infrared Detection System AN/AAS-36

b. Scan Converted Radar

c. P-3C Aircraft

Preparing activity:

Navy - AS

(Project 5821-N153)

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